

DD_RAT-01 - Usage Rating

Version: 1.0 Status: Draft for review Bundle: 05_billing Audience: billing developers, usage rating developers, wallet developers, invoicing developers, QA Related DDs:

- DD_MED-01 - Usage Mediation
- DD_PLM-CFG-07 - Voice Tariff Catalog
- DD_PLM-CFG-02 - Tax Configuration
- DD_BIL-05 - Wallet and Topup
- DD_BIL-02-GEN-01 - Invoice Generation
- DD_BIL-02-TAX-01 - Tax Invoice and Gateway
- DD_REP-01 - Reporting Data Mart and Dashboard API

MVP simplification baseline: RAT-01 is offline/batch rating only in this design. Implement after MED-01, wallet/payment basics, and tariff catalog are stable. Do not implement online rating/session control in the first release.

1. Purpose

RAT-01 defines how SOPHIX rates mediated usage records and turns them into billable financial results.

The first required use case is voice usage rating:

```
MED-01 mediated CDR
-> RAT-01 tariff lookup, allowance consumption, tax computation
-> PREPAID wallet debit or POSTPAID rated usage for invoice generation
```

RAT-01 exists because usage charging is different from lifecycle charging. BIL-01 handles lifecycle/business-intent charges such as activation, migration, relocation, fees, and proration. RAT-01 handles high-volume usage records such as CDRs.

2. Ownership Boundary

Concern	Owner
Raw CDR import, normalization, duplicate detection, quarantine	MED-01
Tariff plans, zones, prefixes, rates, allowances definitions	PLM-CFG-07
Usage rating execution, rated usage ledger, allowance consumption	RAT-01
Tax computation for usage charge	PLM-CFG-02 called by RAT-01
PREPAID wallet debit	BIL-05 called by RAT-01
POSTPAID invoice generation from rated usage	BIL-02-GEN-01 reads or consumes RAT-01 rated usage
Tax invoice after payment/topup/invoice	BIL-02-TAX-01
Wallet balance and wallet ledger	BIL-05

Rule: RAT-01 must not parse raw CDR files. It consumes only mediated usage from MED-01.

Rule: RAT-01 must not own wallet balances or invoice numbering. It calls the owning modules.

3. Scope

3.1 In Scope

- Consume UsageMediated events or poll MED-01 records.
- Rate voice calls using PLM-CFG-07 tariff lookup.
- Apply call duration rounding rules.
- Apply charge policy:
- CHARGEABLE;

- ZERO_RATED;
- BLOCKED;
- QUARANTINE.
- Manage per-subscription allowance buckets and allowance consumption.
- Compute tax using PLM-CFG-02.
- Store immutable rated usage records.
- Debit PREPAID wallets via BIL-05.
- Expose POSTPAID rated usage to invoice generation.
- Retry failed rating and failed settlement.
- Quarantine rating failures separately from MED-01 quarantine.

3.2 Out Of Scope

- Raw CDR parsing.
- Destination prefix catalog maintenance.
- Wallet topup handling.
- External payment gateway callbacks.
- Dunning decisions after insufficient wallet balance.
- Invoice PDF generation.
- Tax authority signing.

4. High Level Architecture

```

flowchart LR
    MED["MED-01<br/>Mediated usage"]
    RAT["RAT-01<br/>Usage rating"]
    TAR["PLM-CFG-07<br/>Voice tariff"]
    TAX["PLM-CFG-02<br/>Tax compute"]
    WAL["BIL-05<br/>Wallet debit"]
    INV["BIL-02-GEN-01<br/>Invoice generation"]
    REP["REP-01<br/>Reporting"]

    MED --> RAT
    RAT --> TAR
    RAT --> TAX
    RAT --> WAL
    RAT --> INV
    RAT --> REP

```

5. Core Concepts

5.1 Rated Usage

A rated usage record is the immutable financial result of rating one mediated usage record.

Example:

```

{
  "ratedUsageId": "rated-20260602-000001",
  "mediatedUsageId": "med-20260602-000001",
  "subscriptionId": "SUB-700045",
  "usageType": "VOICE_CALL",
  "zoneCode": "KE_MOBILE",
  "actualDurationSeconds": 75,
  "billableDurationSeconds": 120,
  "grossAmount": 6.00,
  "taxAmount": 0.96,
  "netAmount": 6.96,
  "currency": "KES",
  "settlementStatus": "SETTLED"
}

```

```
}
```

5.2 Allowance Bucket

An allowance bucket tracks remaining included usage for a subscription and cycle.

Example:

```
100 on-net minutes per bill cycle
```

PLM-CFG-07 defines the allowance. RAT-01 creates and consumes the runtime bucket.

5.3 Settlement

Settlement is what happens after rating:

Billing mode	Settlement
PREPAID	Immediate wallet debit through BIL-05.
POSTPAID	Rated usage remains pending for invoice generation.
ZERO_RATED	No settlement, but record is stored for audit/reporting.

6. Runtime Flow

6.1 PREPAID Voice Rating Flow

```
sequenceDiagram
    participant M as MED-01
    participant R as RAT-01
    participant T as PLM-CFG-07
    participant X as PLM-CFG-02
    participant W as BIL-05

    M->>R: UsageMediated event
    R->>T: Voice rating lookup
    R->>R: Apply allowance and duration rounding
    R->>X: Compute tax
    R->>R: Store rated usage
    R->>W: Debit wallet
    W-->>R: Debit result
    R->>R: Mark SETTLED
```

6.2 POSTPAID Voice Rating Flow

```
sequenceDiagram
    participant M as MED-01
    participant R as RAT-01
    participant T as PLM-CFG-07
    participant X as PLM-CFG-02
    participant I as BIL-02-GEN-01

    M->>R: UsageMediated event
    R->>T: Voice rating lookup
    R->>R: Apply allowance and duration rounding
    R->>X: Compute tax
    R->>R: Store rated usage as POSTPAID_PENDING_INVOICE
    I->>R: Read rated usage for cycle
    I->>I: Generate invoice line summary
```

7. Business Rules

Rule ID	Rule	Notes
R-RAT-01	Only mediated records from MED-01 can be rated.	No raw CDR parsing.
R-RAT-02	Rating is idempotent on mediated_usage_id.	Same CDR cannot be charged twice.
R-RAT-03	Tariff lookup uses call start time, not processing time.	Historical rates matter.
R-RAT-04	Destination zone and rate come from PLM-CFG-07.	RAT-01 must not hardcode rates.
R-RAT-05	Duration rounding uses the tariff rate's increment fields.	60/60, 1/1, etc.

Rule ID	Rule	Notes
R-RAT-06	Allowance consumption happens before paid rating.	Free seconds reduce billable paid seconds.
R-RAT-07	Allowance consumption must be atomic per subscription/cycle/allowance.	Prevent concurrent over-consumption.
R-RAT-08	Tax is computed by PLM-CFG-02 for non-zero gross charge.	Zero-rated records have zero tax.
R-RAT-09	PREPAID settlement calls BIL-05 with a stable idempotency key.	Prevent double debit.
R-RAT-10	POSTPAID rated usage remains immutable and is consumed by invoice generation once.	Prevent duplicate invoice lines.
R-RAT-11	Insufficient prepaid wallet balance marks rating settlement failed, not rating failed.	Rating amount is still valid.
R-RAT-12	Unknown tariff/rate sends record to rating quarantine.	MED was valid, rating config is missing.
R-RAT-13	Corrections after rating require reversal/adjustment flow, not editing rated usage.	Audit protection.

8. Data Model

8.1 usage_rating_job

Tracks event or batch processing jobs.

Use this table to monitor rating worker progress.

Column	Type	Purpose	Sample
rating_job_id	UUID PK	Job id.	urj-001
operator_code	TEXT	Operator/tenant.	WIK
job_type	TEXT	EVENT, BATCH, REPLAY.	EVENT
source_ref_id	TEXT	Event id, batch id, or replay id.	evt-usage-mediated-001
status	TEXT	RUNNING, COMPLETED, FAILED, PARTIAL.	COMPLETED
records_processed	INTEGER	Count processed.	1
records_rated	INTEGER	Count rated.	1
records_quarantined	INTEGER	Count quarantined.	0
started_at	TIMESTAMP	Start time.	2026-06-02T11:06:00Z
completed_at	TIMESTAMP NULL	Completion time.	2026-06-02T11:06:01Z

Recommended indexes:

- (operator_code, status, started_at).

8.2 rated_usage_record

Stores one immutable rated usage result.

Use this table as the source of truth for usage charge audit and postpaid invoice generation.

Column	Type	Purpose	Sample
rated_usage_id	UUID PK	Rated usage id.	rated-001
mediated_usage_id	TEXT UNIQUE	MED-01 record id.	med-20260602-000001
operator_code	TEXT	Operator/tenant.	WIK
usage_type	TEXT	VOICE_CALL.	VOICE_CALL
billing_mode	TEXT	PREPAID, POSTPAID.	PREPAID
subscription_id	TEXT	Subscription.	SUB-700045
customer_id	TEXT	Customer.	CUS-100045
account_id	TEXT NULL	Billing account.	ACC-900045
service_ref	TEXT	PLM service.	svc_VOIP_STD

Column	Type	Purpose	Sample
package_ref	TEXT NULL	Package at usage time.	pkg_INET_VOICE_STD
tariff_plan_code	TEXT	Applied tariff plan.	WIK_RES_VOICE_STD_2026
zone_code	TEXT	Destination zone.	KE_MOBILE
time_band_code	TEXT	Time band.	ANYTIME
actual_duration_seconds	INTEGER	Source duration.	75
allowance_seconds_applied	INTEGER	Seconds paid by allowance.	0
billable_duration_seconds	INTEGER	Charged seconds after allowance and rounding.	120
unit_price	DECIMAL(18,6)	Applied unit price.	3.000000
gross_amount	DECIMAL(18,6)	Amount before tax.	6.000000
tax_amount	DECIMAL(18,6)	Tax total.	0.960000
net_amount	DECIMAL(18,6)	Gross plus tax.	6.960000
currency_code	TEXT	Currency.	KES
tax_breakdown_json	JSONB	Tax lines from PLM-CFG-02.	[{"taxType": "VAT", "amount": 0.96}]
rating_status	TEXT	RATED, ZERO_RATED, QUARANTINED.	RATED
settlement_status	TEXT	NOT_REQUIRED, PENDING, SETTLED, FAILED, POSTPAID_PENDING_INVOICE, INVOICED.	SETTLED
settlement_ref_id	TEXT NULL	Wallet transaction id or invoice id.	wtx-001
rated_at	TIMESTAMP	Rating time.	2026-06-02T11:06:00Z

Recommended indexes:

- Unique (mediated_usage_id).
- (operator_code, settlement_status, rated_at).
- (subscription_id, rated_at).
- (billing_mode, settlement_status, rated_at).

8.3 usage_allowance_bucket

Stores runtime allowance balance for one subscription and cycle.

Use this table when a tariff plan includes free minutes.

Column	Type	Purpose	Sample
bucket_id	UUID PK	Bucket id.	uab-001
operator_code	TEXT	Operator/tenant.	WIK
subscription_id	TEXT	Subscription.	SUB-700045
tariff_plan_code	TEXT	Tariff plan.	WIK_RES_VOICE_STD_2026
allowance_code	TEXT	PLM-CFG-07 allowance.	ONNET_100_MIN
cycle_ref	TEXT	Billing or allowance cycle.	2026-06
included_seconds	INTEGER	Initial included seconds.	6000
consumed_seconds	INTEGER	Used seconds.	120
remaining_seconds	INTEGER	Remaining seconds.	5880
status	TEXT	ACTIVE, EXHAUSTED, EXPIRED.	ACTIVE
created_at	TIMESTAMP	Creation timestamp.	2026-06-01T00:00:00Z
updated_at	TIMESTAMP	Last update timestamp.	2026-06-02T11:06:00Z

Recommended indexes:

- Unique (operator_code, subscription_id, allowance_code, cycle_ref).
- (subscription_id, status).

8.4 usage_allowance_consumption

Stores allowance consumption per rated usage record.

Use this table for audit and customer usage history.

Column	Type	Purpose	Sample
consumption_id	UUID PK	Consumption id.	uac-001
bucket_id	UUID FK	Allowance bucket.	uab-001
rated_usage_id	UUID FK	Rated usage.	rated-001
consumed_seconds	INTEGER	Seconds consumed.	60
created_at	TIMESTAMP	Creation timestamp.	2026-06-02T11:06:00Z

Recommended indexes:

- Unique (bucket_id, rated_usage_id).

8.5 usage_rating_quarantine

Stores mediated records that cannot be rated due to rating configuration or settlement preparation errors.

Use this table when MED-01 was valid but rating cannot complete.

Column	Type	Purpose	Sample
rating_quarantine_id	UUID PK	Quarantine id.	urq-001
mediated_usage_id	TEXT	MED-01 record id.	med-20260602-000002
operator_code	TEXT	Operator/tenant.	WIK
reason_code	TEXT	Why rating failed.	VOICE_RATE_NOT_FOUND
severity	TEXT	LOW, MEDIUM, HIGH.	HIGH
status	TEXT	OPEN, IN_REVIEW, RESOLVED, IGNORED.	OPEN
context_json	JSONB	Rating context.	{"calledNumber":"+998..."}
resolution_notes	TEXT NULL	Repair notes.	Tariff prefix added
created_at	TIMESTAMP	Creation timestamp.	2026-06-02T11:06:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	NULL

Recommended indexes:

- (operator_code, status, reason_code, created_at).

8.6 usage_settlement_attempt

Stores each attempt to settle a rated usage record.

Use this table for wallet debit retries and postpaid invoice handoff audit.

Column	Type	Purpose	Sample
settlement_attempt_id	UUID PK	Attempt id.	usa-001
rated_usage_id	UUID FK	Rated usage.	rated-001
target_module	TEXT	BIL-05 or BIL-02-GEN-01.	BIL-05
target_endpoint	TEXT	Endpoint called.	/api/wallets/WAL-200045/debit
idempotency_key	TEXT	Settlement idempotency.	RAT:med-20260602-000001
status	TEXT	SUCCESS, FAILED_RETRYABLE, FAILED_FINAL.	SUCCESS
request_payload_redacted	JSONB	Redacted request.	{"amount":6.96}
response_payload_redacted	JSONB NULL	Redacted response.	{"walletTransactionId":"wtx-001"}
error_code	TEXT NULL	Error if failed.	INSUFFICIENT_BALANCE
created_at	TIMESTAMP	Attempt time.	2026-06-02T11:06:01Z

Recommended indexes:

- (rated_usage_id, created_at).
- (status, created_at).

9. APIs

9.1 Rating APIs

```
POST /api/usage-rating/rate
POST /api/usage-rating/rate-batch
GET  /api/usage-rating/rated-usages
GET  /api/usage-rating/rated-usages/{rated_usage_id}
POST /api/usage-rating/rated-usages/{rated_usage_id}/retry-settlement
```

9.2 Postpaid Invoice Read APIs

Used by invoice generation.

```
GET  /api/usage-rating/postpaid-usage?subscriptionId={id}&cycleRef={cycle}
POST /api/usage-rating/postpaid-usage/mark-invoiced
```

mark-invoiced must include the invoice id and the list of rated usage ids included in the invoice.

9.3 Quarantine APIs

```
GET  /api/usage-rating/quarantine?status=OPEN
POST /api/usage-rating/quarantine/{rating_quarantine_id}/replay
POST /api/usage-rating/quarantine/{rating_quarantine_id}/ignore
```

10. Events

Event	Topic	When emitted
UsageRated	sophix.usage.rated	A usage record is rated.
UsageRatingQuarantined	sophix.usage.rating.quarantined	Rating cannot complete.
PrepaidUsageDebited	sophix.usage.prepaid.debited	Wallet debit succeeds.
PrepaidUsageDebitFailed	sophix.usage.prepaid.debitfailed	Wallet debit fails.
PostpaidUsageReadyForInvoice	sophix.usage.postpaid.readyforinvoice	POSTPAID rated usage is available.
PostpaidUsageInvoiced	sophix.usage.postpaid.invoiced	Invoice generation marked rated usage invoiced.

Sample UsageRated:

```
{
  "eventType": "UsageRated",
  "operatorCode": "WIK",
  "ratedUsageId": "rated-20260602-000001",
  "mediatedUsageId": "med-20260602-000001",
  "subscriptionId": "SUB-700045",
  "usageType": "VOICE_CALL",
  "zoneCode": "KE_MOBILE",
  "grossAmount": 6.00,
  "taxAmount": 0.96,
  "netAmount": 6.96,
  "currency": "KES",
  "settlementStatus": "SETTLED"
}
```

11. Frontend Usage

FE-APP-01 Backoffice uses RAT-01 for:

- Rated usage search by customer, subscription, date, usage type, and status.
- Rating quarantine queue.
- Settlement failure queue.

- Postpaid usage ready-for-invoice checks.
- Customer/subscription rated usage history.
- Allowance bucket inspection.

Self-care should read customer-safe usage summaries from a read API or reporting layer, not from raw rating operational tables.

12. Infra Recommendation

12.1 Medium Budget Deployment

Component	Recommendation
API pods	2 pods, 1 vCPU, 1.5 GB RAM each.
Rating workers	1-2 workers, 2 vCPU, 4 GB RAM each.
Settlement workers	1 worker, 1 vCPU, 2 GB RAM.
DB	Billing PostgreSQL schema usage_rat; partition rated usage monthly.
Cache	Redis/local cache for tariff lookup and tax config, invalidated by catalog events.
Kafka	Consume UsageMediated, emit rating/settlement events.

12.2 Recommended Partitions

Partition monthly:

- rated_usage_record
- usage_settlement_attempt
- usage_rating_quarantine
- usage_allowance_consumption

Allowance bucket table is current-state plus cycle history and does not need partitioning at launch.

12.3 OS Recommendation

Use Ubuntu Server 24.04 LTS for VPS deployments or hardened Debian/Ubuntu slim base images for containers.

Rating workers should run near the database and cache. They do not need public internet exposure.

13. AWS Deployment Recommendation

AWS component	Usage
ECS/EKS	Run rating API and workers.
RDS PostgreSQL	Rated usage, allowance, quarantine, settlement tables.
MSK or managed Kafka equivalent	UsageMediated consumption and rating events.
ElastiCache Redis	Tariff/tax lookup cache and short locks.
CloudWatch	Worker lag, rating failures, settlement failures.

Required alarms:

- Rating worker backlog growing.
- Rating quarantine count above threshold.
- Prepaid debit failure spike.
- Postpaid usage not invoiced after cycle close.
- Tariff lookup errors after catalog change.

14. Testing Requirements

14.1 Unit Tests

- 75 seconds with 60/60 increments becomes 120 billable seconds.
- 61 seconds with 1/1 increments remains 61 billable seconds.
- Zero-rated rate creates zero gross/tax/net.
- Allowance reduces paid seconds.
- Allowance bucket lock prevents over-consumption.
- Tax compute response is stored in rated usage.
- Duplicate mediated usage returns prior rated result.

14.2 Integration Tests

- UsageMediated event rates successfully.
- PLM-CFG-07 missing rate creates rating quarantine.
- PREPAID rated usage debits BIL-05 exactly once.
- BIL-05 insufficient balance marks settlement failed.
- POSTPAID rated usage is returned to invoice generation and marked invoiced.
- Retry settlement uses same idempotency key.

14.3 End To End Tests

- Raw CDR from MED-01 becomes wallet debit for PREPAID customer.
- Raw CDR from MED-01 becomes invoice line source for POSTPAID customer.
- Duplicate CDR does not create second wallet debit.
- Tariff update effective tomorrow does not change today's CDR rating.

15. Definition Of Done

RAT-01 is complete when:

- It consumes mediated usage from MED-01.
- It looks up tariffs through PLM-CFG-07.
- It applies duration rounding and allowances.
- It computes tax through PLM-CFG-02.
- It stores immutable rated usage records.
- PREPAID settlement debits BIL-05 idempotently.
- POSTPAID rated usage can be consumed once by invoice generation.
- Rating quarantine and replay are implemented.
- Settlement retry is implemented.
- Backoffice can inspect rated usage, allowance buckets, quarantine, and settlement failures.